

Objectives

- To demonstrate the working of process scheduling algorithms.

Pre-Lab Theory:

First come First serve algorithm:

```
#include<stdio.h>
```

```
main()
{
    int bt[20], wt[20], tat[20], i, n;
    float wtsum, tatsum;
    //clrscr();
    printf("\nEnter the number of processes -- ");
    scanf("%d", &n);
    for(i=0;i<n;i++)
    {
        printf("\nEnter Burst Time for Process %d -- ", i);
        scanf("%d", &bt[i]);
    }
    wt[0] = wtavg = 0;
    tat[0] = tatavg = bt[0];
    for(i=1;i<n;i++)
    {
        wt[i] = wt[i-1] +bt[i-1];
        tat[i] = tat[i-1] +bt[i];
        wtsum = wtsum + wt[i];
        tatsum = tatsum + tat[i];
    }
    printf("\t PROCESS \tBURST TIME \t WAITING TIME\t TURNAROUND\n");
    for(i=0;i<n;i++)
    printf("\n\t P%d \t\t %d \t\t\t %d \t\t\t %d", i, bt[i], wt[i], tat[i]);
    printf("\nAverage Waiting Time -- %f", wtsum/n);
```

```
printf("\nAverage Turnaround Time -- %f", tatsum/n);}
```

In-Lab Tasks:

Task1

Add arrival times to the FCFS Algorithm code given in the prelab to implement the complete FCFS algorithm in c language.

Brief the working/output:

Task2

Implement Shortest Job First algorithm in c programming language.

Brief the working/output:

Task3

Implement the Shortest Remaining Time First algorithm in c programming language.

Brief the working/output:

Task4

Implement the Round Robin algorithm in c programming language

Brief the working/output:

LAB # 8 process scheduling algorithms

Table 1: Revise statements of LAB rubric for assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed all assigned tasks independently, strictly followed all health and safety protocols, and presented the results clearly and accurately.	4	
Good	The student completed the assigned tasks with minimal guidance, mostly followed health and safety protocols, and presented the results appropriately.	3	
Average	The student partially completed the assigned tasks, inconsistently followed health and safety protocols, and presented partial or unclear results.	2	
Worst	The student did not complete the assigned tasks, frequently disregarded health and safety protocols, and failed to present results.	1	